Jackknife Ridge Estimation in Linear Mixed Models with Measurement Error

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Abstract:

This study investigates the problem of bias in linear mixed models (LMMs) when ridge estimation is applied in the presence of measurement errors in fixed-effect variables. First, the ridge estimator for LMMs is described, and then a new estimator, called the jackknife ridge estimator (JRE), is proposed. The performance of the proposed estimator is compared with the conventional ridge estimator, demonstrating improvements in terms of bias and mean squared error. In addition, the asymptotic properties of both estimators are derived. Finally, simulation studies and a numerical example are presented to evaluate the efficiency of the proposed estimator.

Keywords: Jackknife ridge; Linear mixed models; Measurement error; Bias reduction; Asymptotic properties

Mathematics Subject Classification (2010): 99X99, 99X99, 99X99.

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1 Introduction

Linear mixed models (LMMs) are widely used for analyzing clustered and correlated data. By incorporating both fixed and random effects, these models provide a flexible framework for modeling complex data structures. In recent years, LMMs have become important tools for analyzing

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longitudinal and repeated measurements data in many fields such as social sciences, economics, and medical studies. Detailed discussions of LMMs can be found in Diggle et al. (1994), Davidian (1995), Zhong et al. (2000), Zare and Rasekh (2011), Demidenko (2013), and Neuhaus and McCulloch (2014). In regression analysis, reliable inference requires several key assumptions, including the accurate measurement of explanatory variables. However, in many practical situations these variables are subject to measurement error, which may lead to biased and inconsistent parameter estimates and misleading conclusions (Stefanski (1985); Fuller (1987)). Measurement error becomes even more challenging in the context of LMMs, since it can distort the interpretation of model parameters. To reduce this bias, approaches such as the corrected score method have been proposed. Zhong et al. (2002) studied this method under normality assumptions and established the consistency and asymptotic normality of the resulting estimators. Another common issue in regression modeling is multicollinearity among explanatory variables, which can cause instability in least squares estimates. Ridge regression, introduced by Hoerl and Kennard (1970), is a well-known technique for addressing this problem by stabilizing parameter estimation through regularization. Several studies have extended ridge estimation to more complex models. For example, Emami (2018) considered ridge estimation in semiparametric models with measurement error, while Özkale and Can (2017) and Yavarizadeh et al. (2022) investigated ridge-based approaches within LMMs. Although biased estimators such as ridge regression are often used to handle multicollinearity, reducing bias remains an important objective. The Jackknife method, originally proposed by (Quenouille (1949),Quenouille (1956)), is a well-known technique for bias reduction and variance estimation. This approach has been applied in various statistical contexts, including regression diagnostics Cook (1977) and ridge-type estimators (Nomura (1988); Nyquist (1988); Özkale and Kuran (2019)). Motivated by these developments, this paper proposes a jackknife ridge estimator for linear mixed models in the presence of measurement error. The proposed method aims to improve estimation accuracy by reducing bias while maintaining favorable mean squared error properties. The remainder of the paper is organized as follows. Section 2 presents the model formulation and estimation procedures. Section 3 introduces the proposed estimator and studies its asymptotic properties. Section 4 reports simulation results, and Section 5 provides a numerical example. Finally, Section 6 concludes the paper.

2 Model Definition and Estimation

Consider the linear mixed measurement error model:

$$y = Z\beta + Ub + \varepsilon, \quad X = Z + \Delta, \tag{2.1}$$

where y is an $n \times 1$ vector of observations, β is a $p \times 1$ vector of fixed-effects parameters, and $U = [U_1| \dots |U_w]$ is a known design matrix. The random effects are $b \sim N(0, \sigma^2\Sigma)$, where Σ is a block diagonal matrix with blocks $\gamma_i I_{q_i}$ and $\gamma_i = \sigma_i^2/\sigma^2$. The unobservable Z (rank p) is measured as $X = Z + \Delta$, with $\Delta \sim MN(0, I_n \otimes \Lambda)$, where Λ is a known $p \times p$ matrix. Errors $\varepsilon \sim N(0, \sigma^2 I_n)$ are independent of b and Δ.

To implement the Jackknife method, we transform (2.1) into its canonical form.Let $V = I_n + U\Sigma U'$ and S be a nonsingular symmetric matrix such that $S'S = V^{-1}$. Pre-multiplying (2.1) by S gives:

$$y^* = Z^*\beta + U^*b + \varepsilon^*, \quad X^* = Z^* + \Delta^*, \tag{2.2}$$

where $y^* = Sy$, $Z^* = SZ$, and $\mathrm{var}(y^*) = \sigma^2 I_n$. Given the symmetry of $Z^{*'}Z^* = Z'V^{-1}Z$, there exists an orthogonal matrix Q such that $Q'Z'V^{-1}ZQ = \Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$. The canonical model becomes:

$$y^* = Z^{**}\alpha + \varepsilon^*, \quad X^{**} = Z^{**} + \Delta^{**}, \tag{2.3}$$

where $Z^{**} = Z^*Q$, $\alpha = Q'\beta$, and $Z^{**'}Z^{**} = \Lambda$.

The ridge estimates for α, σ^2, and b (assuming known γ) are derived by differentiating the penalized likelihood functions:

$$\hat{\alpha}_k = (X^{**'}X^{**} + kI_p)^{-1}X^{**'}y^*, \quad \hat{\sigma}_k^2 = \frac{1}{n}(y'V^{-1}y - \hat{\beta}_k'X'V^{-1}y), \quad \hat{b}_k = \Sigma U'V^{-1}(y - X\hat{\beta}_k). \tag{2.4}$$

For unknown variance components γ_i, we use the iterative ridge estimator:

$$\hat{\sigma}_{ki}^2 = \frac{[\hat{b}_{ki}'\hat{b}_{ki} - \mathrm{tr}(\hat{D}_{ki}'\hat{D}_{ki})\hat{\beta}_k'\Lambda\hat{\beta}_k]}{q_i - \mathrm{tr}(T_{ii})}, \quad \hat{D}_{ki} = \gamma_{ki}U_i'\hat{V}_k^{-1}, \tag{2.5}$$

where T_{ii} is the ii-th block of matrix $T = (I_q + U'V^{-1}U\Sigma)^{-1}$.

3 Jackknife Estimator in Linear Ridge Mixed Measurement Error Models and Asymptotic Properties

The Jackknife method provides a robust framework for bias reduction. Let $\hat{\alpha}_{k(-i)}$ denote the ridge estimator obtained by excluding the i-th observation. Using the Sherman-Morrison-Woodbury theorem, we approximate the jackknife ridge estimator (JRE) as:

$$\hat{\alpha}_{k(J)} \approx \hat{\alpha}_k + (\mathbf{X}^{**\prime}\mathbf{X}^{**} - \mathrm{tr}(\mathbf{V}^{-1})\mathbf{\Lambda}^{**} + k\mathbf{I}_p)^{-1} \sum_{i=1}^{n} \mathbf{x}_i^{**}(y_i^* - \mathbf{x}_i^{**'}\hat{\alpha}_k), \tag{3.1}$$

where $\mathbf{x}_i^{**'}$ is the i-th row of $\mathbf{X}^{**}$. By simplifying the weighted pseudo-values $\Theta_i = \hat{\alpha}_k + n(1 - w_{ii})(\hat{\alpha}_k - \hat{\alpha}_{k(-i)})$, the JRE of α is derived as:

$$\hat{\alpha}_{k(J)} \approx [\mathbf{I}_p - k^2(\mathbf{X}^{**\prime}\mathbf{X}^{**} - \mathrm{tr}(\mathbf{V}^{-1})\mathbf{\Lambda}^{**} + k\mathbf{I}_p)^{-2}]\hat{\alpha}_k. \tag{3.2}$$

Given $\alpha = \mathbf{Q}'\beta$ and $\mathbf{X}^{**'}\mathbf{X}^{**} = \mathbf{Q}'(\mathbf{X}^{*'}\mathbf{X}^*)\mathbf{Q}$, the jackknife ridge estimate for the fixed-effects parameter β is:

$$\hat{\beta}_{k(J)} \approx [\mathbf{I}_p - k^2(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X} - \mathrm{tr}(\mathbf{V}^{-1})\mathbf{\Lambda} + k\mathbf{I}_p)^{-2}]\hat{\beta}_k, \tag{3.3}$$

where $\hat{\beta}_k = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X} - \mathrm{tr}(\mathbf{V}^{-1})\mathbf{\Lambda} + k\mathbf{I}_p)^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y}$. To investigate the asymptotic behavior of the proposed JRE, we rely on the normality of $\xi = n^{-1/2}X^{*'}y^* = n^{-1/2}X'V^{-1}y$. Assuming the existence of the limits for $g = n^{-1}(Z^{*'}Z^*)$ and $D = n^{-1}(Z^{*'}Z^* + kI_p)$, we establish the following theorem:

As $n \to \infty$, the estimator $\hat{\beta}_{k(J)}$ follows an asymptotic normal distribution, $\sqrt{n}(\hat{\beta}_{k(J)} - \beta) \xrightarrow{d} N(0, \mathrm{Avar}(\hat{\beta}_{k(J)}))$, where the asymptotic covariance matrix is given by:

$$\mathrm{Avar}(\hat{\beta}_{k(J)}) = g^{-1}(B + \sigma^2 M^{-1})g^{-1} + k^4 D^{-2} g^{-1}(B + \sigma^2 M^{-1})g^{-1}D^{-2}, \tag{3.4}$$

with $M = (Z^{*'}Z^*)^{-1}$ and $B = [\mathrm{tr}(V^{-1})\sigma^2 + \beta'(Z'V^{-2}Z)\beta]\Lambda$.

Proof Sketch: Using the property $\mathrm{var}(\xi) = n^{-1}(B+\sigma^2 M^{-1})$, we expand $\sqrt{n}\hat{\beta}_{k(J)}$ around β using a Taylor series expansion. Since ξ is asymptotically normal, the asymptotic variance of $\sqrt{n}\hat{\beta}_{k(J)}$ is obtained via the delta method by propagating the variance of ξ through the transformation defining the estimator.

Furthermore, the asymptotic properties of the standard ridge estimator and the corrected score estimator (CSE) can be recovered as special cases. Specifically, the ridge estimator $\hat{\beta}_k$ has an asymptotic normal distribution with mean $M_k M^{-1}\beta$ and covariance $M_k(B + \sigma^2 M^{-1})M_k$. Similarly, the CSE estimator $\hat{\beta}$ is asymptotically normal with mean β and covariance $M(B + \sigma^2 M^{-1})M$. These results facilitate a direct performance comparison between the proposed JRE and existing estimation methods under measurement error conditions.

4 Simulation Study

We conducted a Monte Carlo simulation (1000 replications) to evaluate the finite-sample performance of the proposed JRE compared to the ridge estimator and CSE. We considered factors including sample size (n), measurement error variance (Λ), and cluster variance components (σ^2, σ_1^2). The performance was assessed using the Estimated Mean Square Error (EMSE) and Absolute Bias (ABIAS).

Table 1: Simulated MSE values of CSE ($\hat{\beta}$), Ridge ($\hat{\beta}_k$), and JRE ($\hat{\beta}_{k(J)}$).

			$\rho = 0.70$			$\rho = 0.85$		
Λ	(σ^2, σ_1^2)	n	$MSE(\hat{\beta})$	$MSE(\hat{\beta}_k)$	$MSE(\hat{\beta}_{k(J)})$	$MSE(\hat{\beta})$	$MSE(\hat{\beta}_k)$	$MSE(\hat{\beta}_{k(J)})$
		30	0.1840	0.2048	0.1849	0.1781	0.1849	0.1764
0.01	(0.5,0.8)	60	0.1411	0.1562	0.1417	0.1247	0.1333	0.1247
		90	0.1320	0.1431	0.1323	0.1109	0.1179	0.1110
		30	0.2203	0.2412	0.2206	0.2116	0.2130	0.2074
0.01	(1,2)	60	0.1542	0.1730	0.1550	0.1401	0.1495	0.1397
		90	0.1419	0.1567	0.1425	0.1197	0.1280	0.1197
		30	0.1830	0.2016	0.1836	0.1769	0.1803	0.1748
0.04	(0.5,0.8)	60	0.1379	0.1522	0.1384	0.1231	0.1309	0.1231
		90	0.1284	0.1390	0.1286	0.1086	0.1152	0.1087
		30	0.2211	0.2389	0.2208	0.2157	0.2135	0.2109
0.04	(1,2)	60	0.1517	0.1693	0.1524	0.1394	0.1475	0.1389
		90	0.1388	0.1528	0.1392	0.1180	0.1256	0.1180
		30	0.1825	0.1972	0.1825	0.1816	0.1804	0.1788
0.09	(0.5,0.8)	60	0.1329	0.1457	0.1332	0.1210	0.1272	0.1208
		90	0.1225	0.1323	0.1228	0.1051	0.1107	0.1051
		30	0.2239	0.2361	0.2226	0.2253	0.2161	0.2151
0.09	(1,2)	60	0.1480	0.1635	0.1484	0.1389	0.1447	0.1382
		90	0.1337	0.1466	0.1341	0.1157	0.1219	0.1155

As evidenced by the results in Tables 1 and 2, the proposed JRE consistently yields smaller EMSE and ABIAS values across various parameter combinations. Notably, while the CSE shows significant bias under multicollinearity and measurement error, the JRE successfully mitigates this bias, outperforming the standard ridge estimator. These findings confirm the robustness of our approach in complex linear mixed models.

Table 2: Simulated ABIAS values of CSE ($\hat{\beta}$), Ridge ($\hat{\beta}_k$), and JRE ($\hat{\beta}_{k(J)}$).

			$\rho = 0.70$			$\rho = 0.85$		
Λ	(σ^2, σ_1^2)	n	$ABIAS(\hat{\beta})$	$ABIAS(\hat{\beta}_k)$	$ABIAS(\hat{\beta}_{k(J)})$	$ABIAS(\hat{\beta})$	$ABIAS(\hat{\beta}_k)$	$ABIAS(\hat{\beta}_{k(J)})$
		30	1.1572	1.2000	1.1617	1.2384	1.2690	1.2411
0.01	(0.5,0.8)	60	1.2043	1.2263	1.2054	1.2473	1.2628	1.2479
		90	1.1915	1.2046	1.1920	1.2720	1.2822	1.2722
		30	1.1475	1.2028	1.1548	1.2180	1.2603	1.2230
0.01	(1,2)	60	1.2251	1.2535	1.2270	2.1858	2.1653	2.1547
		90	1.2399	1.2586	1.2407	1.2337	1.2497	1.2343
		30	1.1506	1.1929	1.1549	1.1396	1.1758	1.1428
0.04	(0.5,0.8)	60	1.1986	1.2204	1.1997	1.2427	1.2580	1.2433
		90	1.1860	1.2008	1.1865	1.2678	1.2779	1.2681
		30	1.1406	1.1955	1.1477	1.2123	1.2540	1.2172
0.04	(1,2)	60	1.2195	1.2477	1.2214	2.1902	2.1699	2.1591
		90	1.2348	1.2535	1.2357	1.2291	1.2449	1.2297
		30	1.1390	1.1803	1.1431	1.1286	1.1638	1.1316
0.09	(0.5,0.8)	60	1.1888	1.2101	1.1899	1.2348	1.2497	1.2353
		90	1.1766	1.1911	1.1771	1.2607	1.2705	1.2609
		30	1.1287	1.1826	1.1356	1.2023	1.2431	1.2070
0.09	(1,2)	60	1.2098	1.2376	1.2116	2.1977	2.1778	2.1667
		90	1.2262	1.2446	1.2270	1.2212	1.2367	1.2218

5 Numerical Example

To further validate the performance of the proposed JRE, we applied it to the well-known Boston Housing dataset. Following the setup by Zhong et al. (2002), we analyzed 132 census tracts treated as repeated measurements. The model addressed severe multicollinearity among the fixed-effect variables, as evidenced by a condition number of 354.25 (Ghapani (2019)), and accounted for measurement error in the pollution variable (NOX). Using the estimated ridge parameter $k = 0.0031$, we compared the Ridge Estimator (RE) and the proposed Jackknife Ridge Estimator (JRE).

Table 3: Parameter estimates and performance metrics for the Boston Housing dataset.

Variable	RE	JRE	Variable	RE	JRE
Intercept	8.9724	9.0678	CRIM	-0.0073	-0.0074
RM	-0.0015	-0.0015	CHAS	-0.0282	-0.0281
AGE	0.0011	0.0008	NOX	-0.0095	-0.0103
DIS	0.1082	0.0831	SMSE	0.1626	0.1555
B	0.4517	0.4559	ABIAS	0.1328	0.0015
LSTAT	-0.5431	-0.5358			

The results in Table 3 indicate that the JRE consistently yields lower SMSE and ABIAS values compared to the RE, confirming its superior predictive accuracy and reduced bias.

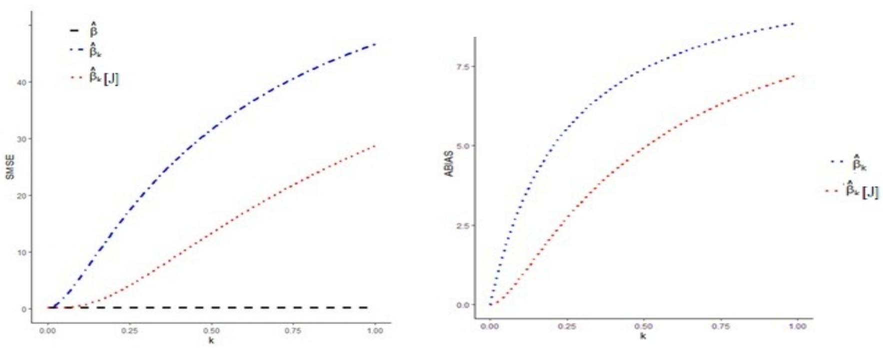

Figure 1: Comparison of SMSE and ABIAS of CSE , Ridge, and JRE estimators versus the ridge parameter k.

Figure 1 illustrates the behavior of the estimators over the interval $k \in [0, 1]$. As shown in this figure, the JRE consistently maintains a smaller SMSE than the standard ridge estimator. Furthermore, the same figure highlights that the JRE significantly reduces the absolute bias (ABIAS) compared to the RE across all values of k. These empirical findings strongly support the theoretical properties established in Section 4, demonstrating that the JRE is an effective alternative for estimation in the presence of multicollinearity and measurement errors.

6 Conclusion

In this study, we proposed the Jackknife Ridge Estimator (JRE) as an effective alternative to the traditional Ridge Estimator (RE) within the framework of linear mixed measurement error models. Given the sensitivity of parameter estimation to both multicollinearity and measurement errors, our approach leverages the jackknife technique to mitigate bias while maintaining a competitive Mean Squared Error (MSE). The theoretical development of the estimator, supported by its asymptotic properties, provides a robust foundation for its application. Furthermore, the results from both the simulation study and the analysis of the Boston Housing dataset consistently demonstrate that the JRE significantly outperforms the standard RE by providing more accurate parameter estimates and lower bias. These findings suggest that the JRE is a reliable tool for practitioners dealing with complex longitudinal data structures subject to measurement error and collinearity issues. Future work may focus on extending this methodology to non-linear mixed models or incorporating robust estimation techniques.

References

Cook, R. D. (1977). Detection of influential observations in linear regression. Technometrics, 19(1) 18-19.

Davidian, M ., Giltian, D.M. (1995). Nonlinear models for repeated measurement data. Chapman and Hall.

Demidenko, E. (2013). Mixed models: theory and applications with R. John Wiley & Sons.

Diggle, P. J., Liang, K. Y., & Zeger, S. L. (1994). Analysis of Longitudinal Data. Clarendon Press.

Emami, H. (2018). Ridge estimation in semiparametric linear measurement error models. Linear Algebra and Its Applications, 552, 127-146.

Fuller, W. A. (1987). Measurement error models. John Wiley & Sons.

Ghapani, F. (2019). Stochastic restricted Liu estimator in linear mixed measurement error models. Communications in Statistics-Simulation and Computation, 51(3): 1220-1233.

Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. Technometrics, 12(1), 55-67.

Neuhaus, J. M., & McCulloch, C. E. (2014). Covariate decomposition methods for longitudinal missing-at-random data. Australian & New Zealand Journal of Statistics, 56(4), 331-345.

Nomura, M. (1988). On the almost unbiased ridge regression estimator. Communications in Statistics-Simulation and Computation, 17(3), 729-743.

Nyquist, H. (1988). Applications of the jackknife procedure in ridge regression. Computational Statistics & Data Analysis, 6(2), 177-183.

Özkale, M. R., & Can, F. (2017). An evaluation of ridge estimator in linear mixed models: an example from kidney failure data. Journal of Applied Statistics, 44(12), 2251-2269.

Özkale, M. R., & Kuran, Ö. (2019). Adaptation of the jackknifed ridge methods to the linear mixed models. Journal of Statistical Computation and Simulation, 89(18), 3413-3452.

Quenouille, M. H. (1949). Approximate tests of correlation in time-series. Mathematical Proceedings of the Cambridge Philosophical Society, 45(3), 483-484.

Quenouille, M. H. (1956). Notes on bias in estimation. Biometrika, 43(3-4), 353-360.

Stefanski, L. A. (1985). The effects of measurement error on parameter estimation. Biometrika, 72(3), 583-592.

Yavarizadeh, B., Rasekh, A., Ahmed, S. E., & Babadi, B. (2022). Ridge estimation in linear mixed measurement error models. Communications in Statistics-Simulation and Computation, 51(6), 3037-3053.

Zare, K., & Rasekh, A. (2011). Diagnostic measures for linear mixed measurement error models. SORT, 35(2), 125-144.

Zhong, X. P., Wei, B. C., & Fung, W. K. (2000). Influence analysis for linear measurement error models. Annals of the Institute of Statistical Mathematics, 52, 367-379.

Zhong, X. P., Fung, W. K., & Wei, B. C. (2002). Estimation in linear models with random effects and errors-in-variables. Annals of the Institute of Statistical Mathematics, 54, 595-606.